MMDU MULLANA OFFICIAL SYLLABUS • BCSE-012 • UNIT 2

Unit 2: Basics of HTML & Website Engineering (Structure, Elements & Mapping)

Subject: Fundamentals of Web Technologies (BCSE-012) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

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1. Website Engineering: The 8-Stage Systems Development Life Cycle (SDLC)

1. Professional Website Engineering Lifecycle (SDLC)

Building enterprise web platforms requires systematic adherence to an 8-stage lifecycle:

SDLC Stage	Engineering Phase	Core Deliverables & Technical Activities
Stage 1	**Planning & Feasibility**	Define Software Requirements Specification (SRS), user personas, scope, and hosting budget.
Stage 2	**UI/UX Wireframing & Design**	Information Architecture (IA) sitemaps, user journeys, and interactive Figma prototypes.
Stage 3	**Frontend Client Engineering**	Semantic HTML5 DOM structure, responsive mobile-first CSS3 layouts, and JS interactivity.
Stage 4	**Backend API & Database**	Server-side business logic (Node.js/Python/Java), SQL/NoSQL schema modeling, and REST APIs.
Stage 5	**QA & Security Testing**	Cross-browser testing, WCAG accessibility audits, OWASP Top-10 security checks, and unit tests.
Stage 6	**Domain & DNS Provisioning**	ICANN registrar domain registration and authoritative DNS zone records (`A` , `AAAA` , `CNAME`).
Stage 7	**CI/CD & Cloud Deployment**	Automated build pipelines (GitHub Actions) deploying to cloud VPS, Kubernetes, or edge CDNs.
Stage 8	**Maintenance & SEO Audits**	Real-time uptime telemetry, security patching, automated DB backups, and Core Web Vitals SEO.

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2. Website Hosting Architectures

1. **Shared Hosting:** Multiple customer websites reside on a single physical server machine, sharing CPU, RAM, and web server software. Low cost, but vulnerable to neighboring resource spikes ("noisy neighbor" problem).

2. **Virtual Private Server (VPS):** A physical server partitioned via hardware hypervisors (KVM) into dedicated virtual slices. Each VPS runs its own isolated operating system kernel with dedicated RAM and CPU guarantees.

3. **Dedicated Server Hosting:** An entire physical computer hardware machine leased exclusively to one organization. Maximum security and compute power; requires dedicated systems administration.

4. **Cloud & Edge CDN Hosting:** Web applications deployed across elastic distributed cloud instances (AWS, Google Cloud) with static assets cached worldwide at edge Point of Presence (PoP) locations for sub-millisecond global delivery.

2. Static vs. Dynamic Websites: Comprehensive 12-Point Analysis

Exhaustive Comparison: Static vs. Dynamic Websites

#	Dimension	Static Website	Dynamic Website
1	**Core Definition**	Displays pre-authored, fixed content stored on disk; appears identical to every visitor.	Generates and personalizes content on-the-fly based on user request, database state, and session.
2	**Technologies Employed**	Pure HTML5, CSS3, and basic vanilla client-side JavaScript.	Server-side languages (PHP, Node.js, Python, Java) paired with database engines.
3	**Database Dependency**	**Zero database requirement.** Text and media paths are hardcoded directly into files.	**Heavily database-driven** (MySQL, PostgreSQL, MongoDB, Oracle).
4	**Server Processing Overhead**	Negligible CPU overhead. Server performs direct disk I/O to socket transmission.	High computation per hit. Server executes business logic and runs SQL queries.
5	**Page Load Speed**	Ultra-fast due to lightweight payload and effortless CDN edge caching worldwide.	Slightly slower initial response time due to server processing and database latency.
6	**Development Complexity**	Very simple and straightforward; accessible to entry-level developers.	High architectural complexity requiring full-stack programming and database schema design.
7	**Hosting & Maintenance Cost**	Minimal to zero cost. Can be hosted free on static platforms (GitHub Pages, Netlify, Vercel).	High recurring cost for application servers, database instances, and auto-scaling compute.
8	**Ease of Content Updating**	Difficult at scale. Modifying a shared header requires editing every individual HTML file.	Effortless. Non-technical users update content via Content Management Systems (CMS / Admin Dashboards).
9	**User Interactivity & Auth**	No user accounts, personalized profiles, shopping carts, or interactive commenting.	Rich interactive capabilities: user authentication, e-commerce checkouts, dynamic filters.
10	**Security Vulnerability Surface**	Practically immune to server-side attacks. No SQL injection or code injection possible.	Susceptible to SQL Injection, Cross-Site Scripting (XSS), CSRF, and session exploits if unpatched.
11	**File Extensions**	Files saved strictly as `.html` or `.htm`.	Files historically saved as `.php`, `.jsp`, `.aspx`, or routed URL endpoints.
12	**Real-World Examples**	Personal portfolios, restaurant menus, product documentation, brochures.	Amazon, Facebook, Netflix, University ERP systems, Net-Banking portals.

3. HTML Fundamental Structure, Core Tags & Text Formatting

1. Fundamental Structure of an HTML Webpage

Every valid modern HTML5 document conforms to the following skeleton:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MMU Engineering Portal</title>
</head>
<body>
  <h1>Department of Computer Science & Engineering</h1>
  <p>Welcome to B.Tech First Year Web Technologies Laboratory.</p>
</body>
</html>
```

Key Document Structure Tags Explained:

- `` **:** Directs the browser rendering engine to parse the document according to modern HTML5 standards, preventing quirks mode.
- `` **:** The root wrapper element; `lang="en"` declares the primary human language for accessibility screen readers.
- `` **:** Houses metadata, external stylesheets, fonts, and scripts not directly painted on the visual canvas.
- `` **:** Specifies Unicode character encoding covering worldwide languages and symbols.
- `` **:** Critical for responsive mobile design; sets page width to match device screen width.
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